

Cloud LearnX

Mathematics

Transformations and Circle Theorems — Practice Worksheet

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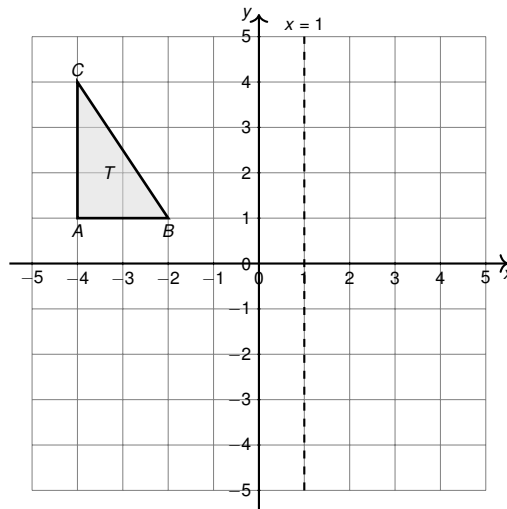
Topic: Transformations and Circle Theorems

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.
Write your answers in the spaces provided.
You must write down all the stages in your working.

1 Triangle T is shown on the grid below.

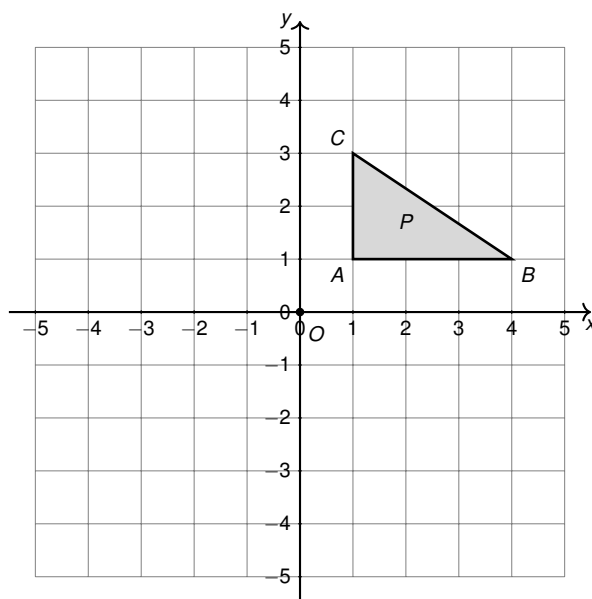


Reflect triangle T in the line $x = 1$.
Label the image T' .

(2)

(Total for Question 1 is 2 marks)

- 2 Triangle P is drawn on the coordinate grid below. A rocket trajectory panel uses triangle P as a marker shape, which must be rotated about the point marked O to plan the next stage of its display.



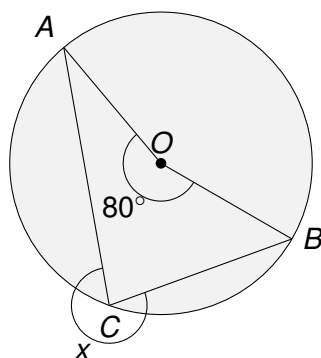
Find the coordinates of the image of triangle P after a rotation of 90° clockwise about the point O .

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(2)

(Total for Question 2 is 2 marks)

3 Points A , B and C lie on the circumference of a circle with centre O , as shown in the diagram.



The angle at the centre, $\angle AOB = 80^\circ$.

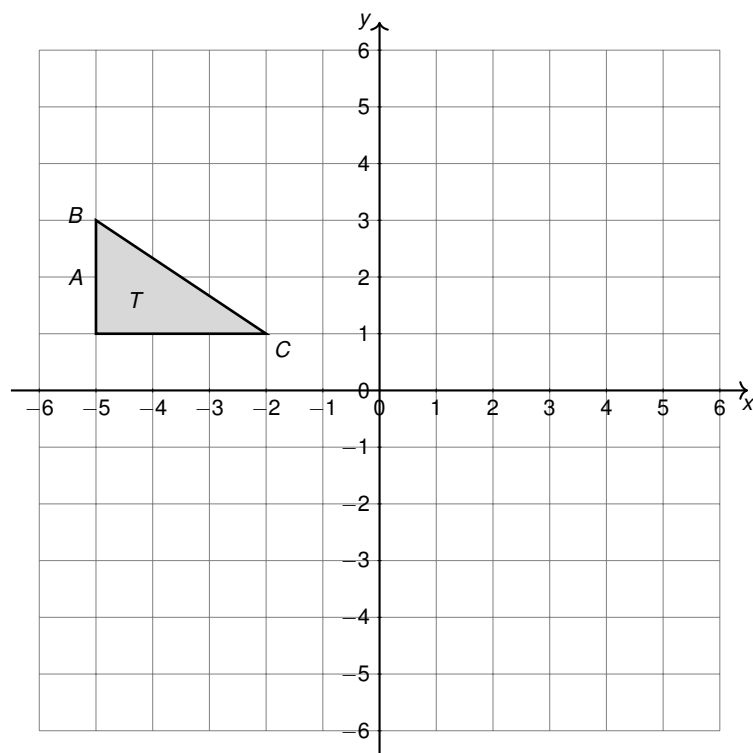
Calculate the size of angle x , marked at the circumference.

$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 3 is 2 marks)

4 Triangle T has vertices $A(-5, 1)$, $B(-5, 3)$ and $C(-2, 1)$.



- (a) Triangle T is translated by the vector $\begin{pmatrix} 7 \\ -4 \end{pmatrix}$ to give triangle U .
Find the coordinates of the vertices of triangle U .

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(1)

- (b) Triangle U is then reflected in the line $x = 0$ to give triangle V .
Describe fully the **single** transformation that maps triangle T directly onto triangle V .

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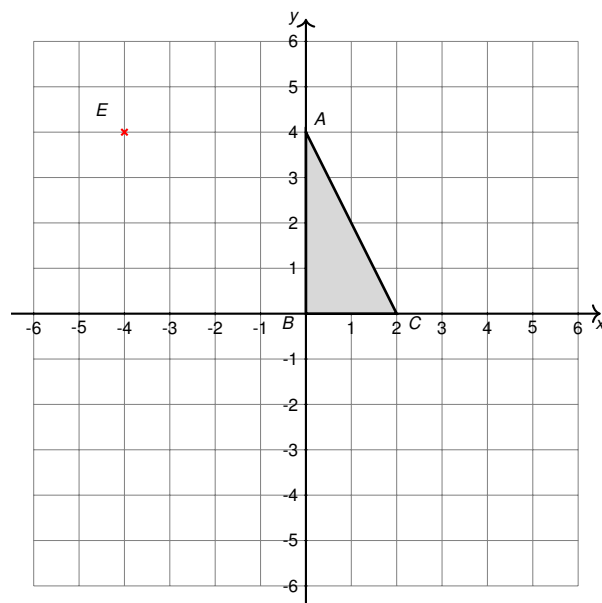
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(2)

(Total for Question 4 is 3 marks)

5 Triangle ABC is drawn on the grid below. Point E marks the centre of enlargement.



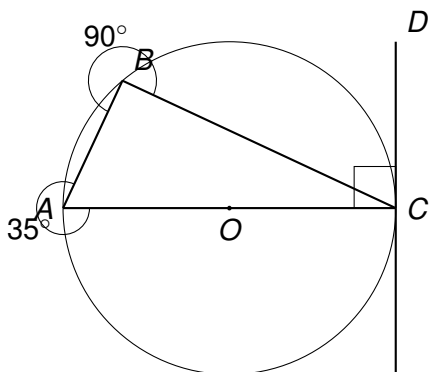
Find the coordinates of the image of triangle ABC after an enlargement with scale factor $\frac{1}{2}$, centre $E(-4, 4)$.

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(3)

(Total for Question 5 is 3 marks)

- 6 A satellite dish engineer models the cross-section of a circular antenna dish as a circle with centre O . Points A , B and C lie on the circumference, where AC is a diameter of the circle. A support strut CD is attached so that it is tangent to the circle at point C .



Angle $BAC = 35^\circ$.

- (a) Show that angle $ABC = 90^\circ$, giving a reason for your answer.

(2)

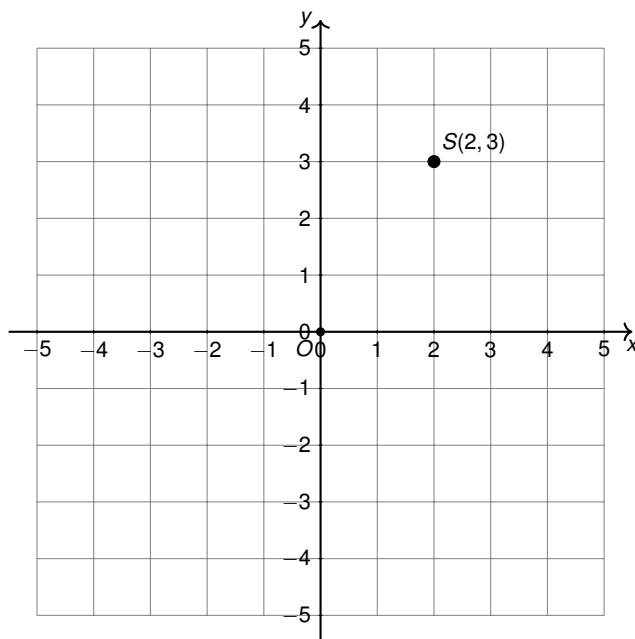
- (b) Work out the size of angle BCD .
Give a reason for each stage of your working.

$BCD = \dots\dots\dots^\circ$

(2)

(Total for Question 6 is 4 marks)

- 7 A space observatory uses a tracking grid to record the position of a satellite as it passes overhead. At a certain moment, the satellite is recorded at point $S(2, 3)$ on the grid, where the origin O represents the observatory's radar dish.



The observatory's software first applies a rotation of 90° clockwise about O to correct for Earth's rotation, giving image point S_1 .

The software then reflects S_1 in the x -axis to align with the second dish, giving the final tracked position S_2 .

- (a) Work out the coordinates of S_1 and S_2 .

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(2)

- (b) Find the single transformation that maps S directly onto S_2 .

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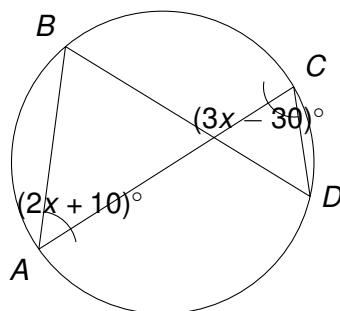
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(2)

(Total for Question 7 is 4 marks)

- 8 A , B , C and D are four points on the circumference of a circle, forming the cyclic quadrilateral $ABCD$.
The angle at A is $(2x + 10)^\circ$ and the angle at C is $(3x - 30)^\circ$.



A historian studying the astrolabe used by medieval astronomers notices that its main dial is divided by a cyclic quadrilateral marked exactly like this one, and wants to check the calibration is correct.

- (a) Show that $x = 40$.

(2)

- (b) Find the size of each of the four angles of the cyclic quadrilateral $ABCD$.

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(2)

(Total for Question 8 is 4 marks)

TOTAL FOR PAPER IS ?? MARKS